

# Anchored multi-omics integration and knowledge-anchored residual discovery

*A never-below-the-best-view integrator, and a tool to find what the known biology misses.*

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## In brief

Combining many kinds of molecular data is widely assumed to improve cancer prediction, yet in practice a single strong data type is hard to beat. This work introduces an *anchored* framework that starts from established biological knowledge — a known high-performing data type, or a fixed textbook gene signature — so the model can never do worse than the existing standard. By analysing only the *residual* signal left after accounting for that anchor, it surfaces orthogonal biological axes that conventional data-fusion overlooks: it recovers a keratinization (basal) program in breast cancer and the field’s established immunotherapy biomarkers in lung cancer. Inverting the idea, a hypothesis can itself be tested against the textbook prior, letting the framework separate genuinely new findings from those already explained by existing literature. It also makes *transportability* explicit — showing how the measurement technology (for example, RNA-seq versus microarray) can make the same biological signal look novel in one cohort and redundant in another. In short, it is a discovery engine that is honest about what is new and what the textbook already knows.

## Abstract

Multi-omics integration is usually sold as “more modalities → better predictions,” yet on real cancer data a strong single modality is hard to beat and naive fusion often does worse. We define an **anchored** integrator that anchors on the empirically strongest (or a fixed prior) modality and adds the others only as a non-negative gated residual, so it is provably never below its best single view and improves on it only where another modality carries orthogonal signal. Across nine TCGA-BRCA endpoints the gate adds nothing where one modality dominates and engages only on a constructed positive control — quantifying, rather than assuming, where multi-omics helps. Generalising the anchor from “best modality” to **established knowledge** (a textbook gene signature, a published clock) yields a zero-parameter prior that nearly matches a fully trained model and gives the programme’s only clean super-additive gain. Finally, gating genome-wide data on a textbook prior and mining the *residual* turns the integrator into a **discovery** tool: it recovers a verified basal/lineage axis beyond “LumA/B = proliferation” (keratinization,  $p \times 10^{-11}$ ) and a verified neuroendocrine/immune axis beyond the ERBB2 amplicon, while correctly finding nothing where the textbook is already complete (ER). Inverting the

frame, a *hypothesis* can itself be expressed as a candidate anchor and tested against the textbook prior; a commonality/mediation decomposition then separates a genuinely novel axis from one that is merely redundant (collinear) or absent, and a five-endpoint  $\times$  four-cohort panel (TCGA RNA-seq, TCGA Agilent, METABRIC, SCAN-B) shows anchor-orthogonal axes transport across platform and cohort while collinear ones do not — their verdict is governed by a nuisance correlation that can flip sign with the assay (RNA-seq vs microarray). The method is modality-agnostic and, above all, honest: discovery where warranted, silence where not, and a quantified account of what fails to reproduce and why.

## 1. Background

The prevailing assumption that adding omics layers improves prediction is contradicted by careful benchmarks: a large TCGA survival study found that adding data types most often *hurt* performance and that mRNA ( $\pm$ miRNA) sufficed for most cancers (Li et al., 2024); late-fusion work shows the dominant modality is task-specific and pan-cancer multi-omics barely exceeds the best single view (Nikolaou et al., 2023); and naive concatenation routinely underperforms structured fusion (Makrodimitris et al., 2023). The honest goal is therefore not “always fuse” but **never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot**. This note defines an integrator built to that contract and then repurposes it as a discovery engine.

## 2. Method

All functions live in `omniomics.multiomics`.

**Anchor selection (`select_anchor`).** Score each modality by repeated stratified-CV performance (primary), tie-break by robustness (subtract a multiple of the CV s.d., weight by coverage). The anchor is the top composite, chosen empirically per task inside the outer CV.

**Gated residual (`anchored_gate` / `anchored_integrate`).** Pin the anchor; add each secondary on the anchor’s residual with a non-negative weight chosen on held-out data,  $= 0$  allowed. The combined score is  $\text{logit}(\text{anchor}) + \beta \cdot \text{secondary}$ .  $= 0$  combined anchor (never below);  $> 0$  only where a secondary earns a margin gain. A constrained, interpretable special case of cooperative learning (Ding et al., 2022).

**Forward gating (`forward_integrate` / `auto_integrate`).** For  $\geq 3$  modalities, add one at a time onto the current residual in rank order, keeping each only if it clears the margin — forward selection over modalities, with an interpretable record of which entered and at what  $\beta$ .

**Knowledge anchor (`knowledge_anchored_integrate` / `signature_score`).** The anchor need not be a data modality: a fixed textbook signature or published

clock (zero trained parameters) becomes the anchor and the data is gated onto its residual. The gate then answers a clinically meaningful question — *does the genome-wide data add anything beyond established biology?*

**Residual discovery (anchored\_residual\_discovery).** Anchor on the prior, gate the data, test that the residual is signal-not-noise, and surface the features driving it that are *orthogonal* to the anchor — candidate biology beyond the textbook — with a matched-random-panel null as the noise control.

Small-sample robustness throughout: a secondary is added only if a non-zero beats the anchor by more than a `gate_margin`, with each 's inner-CV score averaged over repeated splits.

### 3. Results (TCGA-BRCA)

**Anchored integration never falls below the leader.** On LumA/B, RNA alone scores 0.947 and methylation 0.745; a naive stack dips to 0.941 (below RNA) while the gated combiner stays at 0.947 ( $= 0$  in 9/10 repeats). A constructed positive control — a held-out methylation set RNA cannot see — flips this: the gate engages ( $= 4-8$ ) for a significant  $+0.047$ . So the gate is both protective and capable.

**Not RNA-biased.** Run blind on nine expert-defined subtype labels, `auto_integrate` anchors on methylation for all five methylation-defined clusters and on RNA for all four PAM50 calls — anchor selection follows the biology, never below the leader on any of the nine.

**Knowledge anchoring — the first clean super-additive gain.** A 20-gene proliferation index with zero trained parameters scores 0.919 on LumA/B (nearly the trained 1500-gene RNA model, 0.942); gating genome-wide data reaches **0.947 ( $\Delta +0.029$ )**, beating the pure data model. With the Horvath clock as the anchor for normal-tissue age the clock alone (0.947) already beats RNA (0.911) and data adds  $0$  — the textbook suffices and the gate says so.

**Residual discovery, verified.** Mining the proliferation-anchored residual on LumA/B surfaces the basal/ squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial  $r = 0.46$ ). It is verified three ways: a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses (real  $+0.021 >$  all permutations). g:Profiler confirms a coherent program — cornified-envelope formation / keratinization / epidermis development (Reactome  $p = 8 \times 10^{-11}$ ).

**Generalises, reproduces, and discriminates.**

| endpoint        | textbook<br>anchor (0<br>params)      | anchor<br>AUROC | $\Delta$ data adds | verdict                                                                 |
|-----------------|---------------------------------------|-----------------|--------------------|-------------------------------------------------------------------------|
| LumA vs<br>LumB | proliferation<br>index (20<br>genes)  | 0.919           | +0.029             | incomplete<br>→<br>basal/lineage<br>axis<br>(verified)                  |
| HER2 status     | ERBB2<br>amplicon (9<br>genes)        | 0.752           | +0.054             | incomplete<br>→ neuroen-<br>docrine/immune<br>axis (verified,<br>basal) |
| ER status       | ER/luminal<br>signature (20<br>genes) | 0.938           | −0.001             | complete →<br>no hidden<br>axis<br>(specificity)                        |

The HER2 discovery is a second, *different* verified axis (held-out +0.068 in 8/8 splits; panel  $p = 0.024$ ; label-specific), a diffuse neuroendocrine/secretory + immune program distinct from basal. ER is a real-data specificity negative: the textbook signature already matches genome-wide RNA.

**Modality-agnostic.** The discovery runs unchanged on DNA-methylation features; on LumA/B it correctly finds no methylation axis (random and basal-targeted CpGs both  $\Delta = 0$ ) — the basal axis is transcriptional, not predictively methylation-encoded here.

**External validation (independent cohort).** The basal discovery reproduces in METABRIC (microarray,  $n = 1175$  LumA/B): the fixed TCGA-discovered panel adds  $\Delta +0.036$  over the proliferation prior (combined 0.960; beats random panels,  $p = 0.048$ ), and an unbiased re-discovery in METABRIC independently recovers the same basal genes (KRT5/14/17/6B, TP63, SOX10, DSG3/DSC3 ...), overlapping the TCGA panel 20/30 (hypergeometric  $p = 7 \times 10^{-2}$ ). The discovered axis is a reproducible biological signal, not a single-cohort artefact. The HER2 axis, by contrast, does *not* replicate in METABRIC — there the amplicon anchor is near-complete (0.997 vs 0.752 in TCGA), leaving no residual (panel  $\Delta = 0$ , re-discovery overlap 2/30). Reproducibility tracks biological coherence: the pathway-enriched basal axis reproduces; the diffuse HER2 axis was a cohort-specific residual.

**Hypothesis-as-anchor.** The frame also inverts: a *hypothesis* is expressed as a candidate anchor and gated onto the textbook anchor’s residual for a three-way verdict — SUPPORTED (adds beyond the textbook), EXPLAINED\_BY\_TEXTBOOK (predicts alone but redundant once the dominant prior is controlled), or REFUTED — operationalising the observation that

most signatures predict outcome and a hypothesis is novel only if it survives adjustment for the dominant prior. On LumA/B against the proliferation anchor, a basal/keratinization hypothesis is SUPPORTED ( $\Delta +0.039$ ), immune is REFUTED, and a random gene set is EXPLAINED\_BY\_TEXTBOOK (predicts at 0.63 alone but adds 0). Scaled across the 50 MSigDB Hallmark sets, the screen is self-validating — proliferation hallmarks (E2F/G2M/MYC) add exactly 0 beyond the anchor — and, requiring agreement across two proliferation anchors plus FDR, the robust SUPPORTED hits are the estrogen-response lineage programs (matching LumA/B biology). This pathway-level ranking is cohort/platform-sensitive (it does not reproduce in METABRIC microarray, where the anchor-family agreement filter correctly nulls proliferation hallmarks instead), so it is a within-cohort hypothesis-ranking tool, not a cross-platform claim — unlike the gene-level basal discovery, which reproduces. To keep this honest, `hypothesis_anchor_test` returns a gate-free commonality/mediation decomposition (`collinearity_label` {NOVEL, REDUNDANT, INERT}) that separates an *absent* hypothesis from a merely *redundant* (collinear) one: the estrogen-response axis is NOVEL in TCGA (unique  $R^2$  0.038) but REDUNDANT in METABRIC (redundancy 1.00, 96 % mediated through proliferation). A controlled transportability sweep — fixing both marginal effects and varying only the anchor-hypothesis correlation — shows the *same* ER effect is 100 % NOVEL at TCGA’s correlation (+0.19) yet collapses into the collinear/suppression valley at METABRIC’s (−0.17): the verdict is a covariate-distribution (transportability) property, not a change in biology. A five-endpoint panel across four columns (TCGA RNA-seq, TCGA Agilent microarray — the same patients on a different platform — the independent METABRIC microarray cohort, and SCAN-B, a fully independent RNA-seq cohort of ~3,400 tumours) makes the payload concrete: three endpoints transport and two do not. The transportable ones are orthogonal axes (basal→immune and HER2-subtype→immune are NOVEL in all four; ER-status→proliferation carries a small unique slice in all four — robust across platform and independent cohort), while the two non-transporting ones are exactly the ER-collinearity cases (proliferation→ER on Luminal A/B; amplicon→ER on HER2 status). For Luminal A/B the label tracks measurement technology — NOVEL on both RNA-seq cohorts (TCGA RNA-seq, SCAN-B) but REDUNDANT on both microarrays (TCGA Agilent, METABRIC), flipping even on the same TCGA patients. So the caveat attaches specifically to hypotheses whose anchor-collinearity is itself measurement-dependent — anchor-orthogonal axes transport cleanly across platform and cohort.

## 4. Related work

The approach instantiates three established traditions: clinical-offset / incremental-value modelling, where an established model is an offset and omics are selected on the residual (Volkman et al., 2019, a near-exact methodological twin); biologically-informed models that bake known biology into structure (Elmarakeby et al., 2021, P-NET; Liu et al., 2024, Pathformer); and ML on

established signatures (Sarafidis et al., 2024). Our contribution is the packaged, **never-below-anchor, margin-gated** residual on a zero-parameter prior, plus the residual-discovery framing with a matched-random-panel noise control.

## 5. Limitations

Predictive gains over a strong anchor are modest; the method’s value is *routing and discovery*, not large AUROC wins. Genuine super-additive multi-omics gains are rare on real endpoints (one constructed positive control; one real knowledge-anchored gain). The basal discovery is externally validated in METABRIC (replication + independent re-discovery); the HER2 axis tested negative there (cohort-specific — the amplicon anchor is near-complete in METABRIC), so external reproducibility tracks biological coherence. Discovered axes are candidate hypotheses, not causal claims.

## 6. Conclusion

Anchor on the leading view (or the textbook), add the rest as a gated residual, and let the gate tell you — honestly — whether the genome-wide data beats established biology. Where it does, the same residual names the new axis (basal for LumA/B; neuroendocrine/immune for HER2); where it does not (ER), the method stays silent. Discovery where warranted, silence where not.

## References

Li et al., *BMC Med Inform Decis Mak* 2024 · Nikolaou et al., *Cancer Res* 2023 · Makrodimitris et al., *Brief Bioinform* 2023 · Ding, Li, Narasimhan & Tibshirani, *PNAS* 2022 · Volkmann et al., *BMC Med Res Methodol* 2019 · Elmarakeby et al., *Nature* 2021 · Liu et al., *Bioinformatics* 2024 · Sarafidis et al., *ESMO Open* 2024 · Morandini et al., *GeroScience* 2023 · Horvath, *Genome Biology* 2013.

*Code, runners, recorded metrics and CI guards: the omniomics-prototype repository. Full method note: reports/anchored\_integration\_methods.md.*